Let's say we have an index (`all_cars`) distributed among 50 shards and want to resize it to a single-digit shard count: say, 5 shards. Similar to what we did with a splitting operation, the first step is to ensure that our `all_cars` index is read-only, so we set the `index.blocks.write` property to `true`. We can then readjust the shards to a single node. The code in the following listing shows these actions as prerequisites before shrinking the index.

Listing 6.43 Prerequisites before shrinking the index

```
PUT all_cars/_settings
{
  "settings": {
    "index.blocks.write": true,
    "index.routing.allocation.require._name": "node1"
  }
}
```

Now that the source index is all set for shrinking, we can use the shrink operation. The format is as follows: `PUT <source_index>/_shrink/<target_index>`. Let's issue the shrink command to shrink the `all_cars` index.

Listing 6.44 Shrinking the index

```
PUT all_cars/_shrink/all_cars_new
{
  "settings":{
    "index.blocks.write":null,
    "index.routing.allocation.require._name":null,
    "index.number_of_shards":1,
    "index.number_of_replicas":5
  }
}
```

Shrinks the source index

Removes the read-only instruction

Sets the node name to null

Reduces the number of shards

We need to point out a few things in this script. The source index was set with two properties: read-only and the allocation index node name. These settings will be carried over to the new target index if we do not reset them. In the script, we nullify these properties so the target index won't have these restrictions imposed when it's created. We also set the number of shards and replicas on the newly instantiated target index and create a hard link for the target index, pointing to the source index file segments.

> **NOTE** Keep in mind that the number of shards must be smaller than (or equal to) the source index's shard count (after all, we are shrinking the index!). And, yes, as you may have imagined, the target index's shard number must be a factor of the source index's shard number.

While we are here, we can also remove all replicas from the source index so the shrink operation is more manageable. We just need to set the `index.number_of_replicas` property to zero. Remember that the `number_of_replicas` property is dynamic, meaning it can be tweaked on a live index.